

Meeting Notes: Ball-in-Cup Experimental Platform

Two-Rail Human Control Task, Disturbance Adaptation, and Sample-Size Planning

Dipankar Bhattacharya

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1. Overview

This meeting focused on the design of a human experiment built around a ball-in-cup task on a two-axis manual rail platform. The main goals are to study sensorimotor learning, disturbance adaptation, and muscle activity during repeated trials.

Overarching aim. Develop a human-inspired stiffness modulation framework for elbow exosuits that could assist people with Parkinson’s disease and related neurological disorders affecting movement stability and coordination. Healthy human demonstrations on the rail platform provide the target impedance behaviour; a cable-driven manipulator with an antagonistic exosuit replicates and learns that behaviour through a three-phase pipeline (Phase 1: human reference, Phase 2: digital twin + RL, Phase 3: real-robot validation via exo-cable stiffness).

Core idea: the participant manipulates a two-rail platform by hand, while a ball in a cup provides a coupled stabilization task. Platform motion, force/torque sensing, camera-based ball tracking, and EMG are all used to study learning and adaptation.

2. Experimental Platform Concept

The proposed setup consists of two stacked linear rails that provide orthogonal motion:

- the bottom rail moves left and right,
- the upper rail moves up and down,
- a cup with a ball sits on the top stage,
- the participant moves the system using a handle mounted beneath the cup.

A 6-axis force/torque sensor will be placed beneath the cup. Its signals will be calibrated against camera-based measurements of the ball position. The long-term goal is to estimate the ball location from force/torque signals, possibly using a regression-based mapping.

Relevant sensor channels are expected to include:

- F_x and F_y ,
- M_x and M_y ,
- other channels if they improve the regression model.

3. Experimental Blocks

The block protocol has been moved under the phase-first workflow in Section 14.1, specifically under Phase 1 (human rail experiment). This keeps the narrative in execution order: Phase 1 (human rail), Phase 2 (exo + manipulator), and Phase 3 (manipulator-only).

4. Additional Human Measurements

EMG data will be recorded during the trials. This will allow us to examine muscle activation patterns as participants learn the task and respond to perturbations.

Potential questions include:

- whether muscle activation changes with practice,
- whether adaptation is associated with altered co-contraction,
- how EMG patterns change when the disturbance is introduced.

5. Ball Position Estimation and Calibration

Because the ball position is not directly measured by the force sensor, a calibration procedure is needed. The plan is to use synchronized camera tracking and force/torque recordings to build a mapping from sensor outputs to ball position.

A simple workflow is:

1. track the true ball position with a camera,
2. record force/torque data at the same time,
3. fit a regression model to map sensor signals to ball location,
4. validate the estimator on held-out data.

Possible estimation models include linear regression, polynomial regression, or a nonlinear machine-learning model if needed.

6. Sample Size and Participant Number

We also discussed how to determine the number of participants needed. The plan is to use an a priori power analysis.

The sample size depends on:

- the primary outcome measure,
- the main hypothesis or comparison,
- the expected effect size,
- the chosen significance level and power.

A common starting point is $\alpha = 0.05$ and power = 0.80. The participant number can then be estimated once the main behavioral variable is chosen. Candidate primary outcomes include ball-position error, time inside the target circle, number of boundary violations, recovery time after disturbance, or a main EMG metric.

For a within-subject comparison, such as early versus late disturbance trials, a paired-difference analysis can use `TTestPower` in `statsmodels`. A simple approximation is

$$n \approx \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{d_z} \right)^2, \quad (1)$$

where d_z is the standardized effect size for the paired difference, α is the false-positive rate, and $1 - \beta$ is the desired power.

Example: if pilot data suggest that the disturbance increases ball-position error by $\Delta = 4$ units, and the standard deviation of the paired differences is $\sigma_d = 6$ units, then $d_z = \Delta/\sigma_d = 0.67$. Using `statsmodels` with $\alpha = 0.05$ and power = 0.80 gives

$$n \approx 19.5. \quad (2)$$

So the study would need about 20 participants for the minimum estimate, and it would be prudent to recruit a few extra participants to account for unusable trials or dropouts.

The preferred approach is to use a previous similar paper or pilot data to estimate the expected effect size, then compute the required sample size in Python using `statsmodels`. For example, in Python one could write:

```
from statsmodels.stats.power import TTestPower
power_analysis = TTestPower()
nobs = power_analysis.solve_power(effect_size=0.67,
                                  alpha=0.05,
                                  power=0.80,
                                  alternative='two-sided')
```

If prior literature is too weak to support a reliable effect-size estimate, a small pilot study should be run first.

6.1. Participant Numbers Reported in Similar Studies

To anchor our estimate in published practice, the table below summarises participant counts and effect-size hints from comparable cup–ball and complex-object manipulation studies. The local paper folder already contains the first three entries; the others are drawn from related online literature.

Study	N	Notes on design and effect size
Bazzi et al. (2018)	10	Cart–pendulum cup proxy; contraction analysis; large within-subject differences between assistive and resistive perturbation conditions.
Razavian et al. (2021)	4	Pilot-style comparison of four motor-control models.
Razavian et al. (2023)	11	2×2 design (object type $\times$ perturbation), 4 blocks of 100 trials per participant.
Bazzi and Sternad (2020)	10	CCM-based robustness analysis with practice.
Nayeem et al. (2021)	10–12	Preparatory movement and initial conditions for complex object transport.
Hasson et al. (2012)	14	Within-subject manipulation of cup–ball; reported large within-subject practice effects, $d_z \approx 0.7$ –1.0.
Lokesh and Sternad (2024)	12	Underactuated object control; within-subject adaptation effects $d_z \gtrsim 0.6$.
Casadio et al. (2006)	12	Healthy participants on impedance-modulated robotic platform.
Ajoudani et al. (2012)	8	Stiffness transfer from human EMG to manipulator.

What this means for our study. Most cup–ball / complex-object manipulation papers in this line of work use roughly $N = 10$ –15 healthy participants for within-subject designs, and report standardised effect sizes for practice or perturbation effects of approximately $d_z \approx 0.6$ –1.0. Plugging these into the power formula gives:

- $d_z = 0.5$ (small–medium): $n \approx 34$,
- $d_z = 0.67$ (our working example): $n \approx 20$,
- $d_z = 0.8$ (medium–large): $n \approx 15$,
- $d_z = 1.0$ (large, typical of Sternad-lab within-subject practice effects): $n \approx 10$.

A reasonable target range for our protocol is therefore $N \approx 12\text{--}20$ healthy participants, depending on the chosen primary outcome. We would aim for the higher end if the primary outcome is EMG-based (more variable across participants) and the lower end if it is a within-subject kinematic measure with strong practice effects.

7. How the Pieces Connect

We clarified the intended long-term pipeline for the manipulator and exosuit. The human experiments are performed with healthy participants, and their hand motion provides the reference behavior. The robot side then uses a weak or compliant manipulator, with exo joint stiffness mounted on the manipulator, to mimic an “unhealthy” or impaired hand-like interaction.

The connection is easiest to think about in stages:

- The healthy human initially performs the task and provides the reference behavior.
- EMG, rail motion, force/torque, and ball-position data are recorded together.
- On the robot side, the exo is mounted on the manipulator and used to change the joint stiffness.
- The weak or compliant manipulator then mimics the reduced control capacity of an impaired hand.
- A policy is learned from the recorded data, possibly using RL or imitation learning methods.
- That learned policy is then transferred to the manipulator, so the manipulator stiffness and response match the desired interaction behavior.

In this sense, the exo is not the final goal by itself. It acts as an intermediate teaching and stabilization mechanism on the manipulator. It changes the closed-loop dynamics, provides a richer data set, and helps define what a good control strategy looks like before the manipulator is asked to do the same job autonomously.

This also explains the phrase “the manipulator will learn from the exo.” What we mean is that the exo mounted on the manipulator generates a stiffness profile and interaction pattern that can later be distilled into the manipulator controller. The manipulator then becomes the system that executes the learned behavior, while the healthy human data serve as the reference and the exo-on-manipulator trials provide the robot-side adaptation signal.

One practical way to frame the project is:

1. healthy hand control defines the target behavior,
2. exo-mounted manipulator trials collect the robot-side training data,
3. RL or imitation learning learns the policy,
4. the manipulator implements the policy on the rail platform,
5. adaptation is evaluated by comparing performance before and after the exo-informed training.

So the connection is: healthy human behavior provides the template, the exo on the manipulator shapes the stiffness and interaction response, and the manipulator eventually inherits the learned control policy.

Two examples make this clearer.

- **Example 1: smooth tracking.** A healthy participant moves the rail to the right, the ball begins to drift, and the hand makes a small corrective motion to keep the ball near the target. On the robot side, the manipulator is given a weak or compliant joint stiffness, and the exo mounted on the manipulator adjusts that stiffness so the same correction pattern becomes measurable and repeatable. The exo-on-manipulator trial records the resulting response together with EMG and force data. Later, the manipulator is trained to reproduce the same rightward motion plus correction, so it behaves like the healthy hand template.

- **Example 2: disturbance recovery.** A solenoid push perturbs the rail toward the participant. The healthy participant reacts by changing hand force and timing to bring the ball back into the target region. On the robot side, the manipulator starts from a weak stiffness setting, and the exo mounted on the manipulator modulates stiffness to recover the task. RL can use this repeated disturbance-response data to learn a policy that rewards recovery speed and low ball error. The manipulator then uses that learned policy to generate the same compensatory response without the human hand.

In both cases, we are not just copying a path in space. We are trying to copy the interaction rule: how a healthy hand reacts to error, how the robot stiffness changes the response, how quickly it corrects, and how it adapts when the exo or disturbance changes the dynamics.

Another useful way to say it is that the exo mounted on the manipulator can serve as a *teacher* of interaction dynamics. It can make the task easier, more repeatable, or more measurable, which improves the data used for learning. The manipulator is the *student* that later executes the learned stiffness and control behavior on the rail platform.

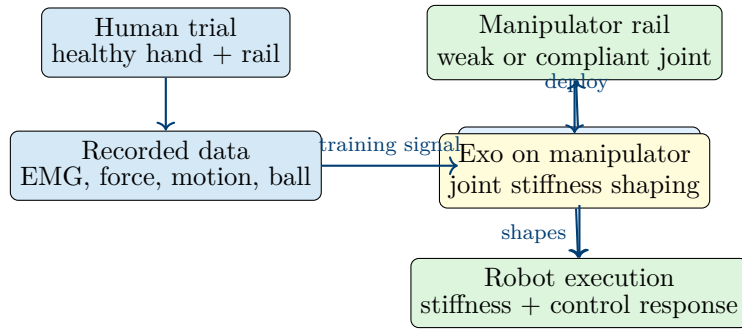


Figure 1: Final experiment diagram. Healthy human rail trials provide the template, and the learned policy is transferred to the manipulator rail with exo-shaped stiffness.

8. Relevant Literature and Design Implications

The closest line of work from Dagmar Sternad’s group is on humans controlling dynamically complex objects, such as cup–pendulum or cup–ball systems. The main message is that humans do not simply move the object along a geometric path. They exploit stability, predictability, impedance, and task dynamics.

Relevant work	Main idea	Implication for our experiment
Bazzi and Sternad (2020); Bazzi et al. (2018)	Humans route trajectories through stable or contracting regions when manipulating complex objects.	Analyze whether healthy participants learn rail motions that make the ball dynamics more predictable and robust.
Razavian et al. (2021); Razavian et al. (2023)	Mechanical impedance and body mechanics strongly shape interaction behavior; fast recovery can arise before neural feedback is available.	Treat exo stiffness on the manipulator as an artificial body-mechanics layer, not just an actuator.
Nayem et al. (2021)	Initial conditions and preparatory movements can simplify interaction with complex objects.	Include a preparatory phase before the rail movement; measure whether participants set up the cup/ball state before moving.
Lokesh and Sternad (2024); Lokesh and Sternad (2025)	Humans adapt to uncertain underactuated object dynamics by improving stability and impedance.	Use uncertainty in ball position or rail disturbance as a controlled difficulty variable.
Nayem et al. (2022)	Functional object interactions can be quantified after stroke.	The weak manipulator can emulate impaired interaction safely before testing patient populations.

This literature suggests that our outcome measures should not be limited to final task success. We should also measure how participants shape interaction dynamics:

- whether ball trajectories become more predictable over trials,
- whether the rail motion enters more stable regions before perturbations,
- whether force/torque signals become smoother or lower variance,
- whether EMG shows reduced co-contraction or better timed activation,
- whether exo-shaped manipulator stiffness reproduces the healthy interaction pattern.

8.1. Online Literature on Impairment Emulation, Impedance Transfer, and RL Stiffness Policies

To ground our specific approach in the wider research community, we surveyed online literature addressing: (i) robotic emulation of paretic or impaired motor capabilities, (ii) imitation-based stiffness or impedance transfer from human/exo to mechanical manipulators, and (iii) reinforcement learning for variable impedance controllers.

- 1. Emulating Motor Impairment on Robotic Testbeds:** In rehabilitation engineering, researchers routinely configure robotic manipulators or haptic interfaces with low gains or active compliant modes to emulate paretic, spastic, or neurologically weak human limbs. For instance, Casadio et al. (2006) and Patinvoh et al. (2020) have programmed robots to mimic the asymmetric force field, muscle weakness, or tremor characteristic of a post-stroke upper limb. This supports our strategy of using a weak or highly compliant joint stiffness setting on the manipulator to mimic the “unhealthy” baseline interaction.
- 2. Human-to-Robot Impedance and Stiffness Transfer:** A well-established paradigm in physical human-robot interaction is the extraction of biological impedance profiles to teach robotic systems how to handle unstable tasks. Seminal work by Ajoudani et al. (2012) and Ajoudani (2018) on the “Tele-impedance” framework demonstrated how human hand stiffness can be estimated online via multi-channel EMG and transferred in real-time to a robot manipulator. Furthermore, Kronander and Billard (2014) and Rozo et al. (2015) developed methods for a robot to learn variable impedance behaviors directly from active human demonstration and physical human-robot interaction (pHRI). In our project, rather than active human guidance, the exo acts as the hardware mediator that physically enforces the targeted impedance

profile on the weak manipulator. The manipulator then “learns from the exo” by digesting these physically assisted interaction trials into a parametric impedance controller.

3. **Reinforcement Learning for Variable Impedance Control (VIC-RL):** Our proposed policy learning step (using RL to discover optimal stiffness behaviors) matches the Variable Impedance Control via Reinforcement Learning line of research pioneered by [Buchli et al. \(2011\)](#) and extended by [Bogdanovic et al. \(2020\)](#). In these papers, the RL action space represents virtual joint or Cartesian stiffness matrix parameters (K, B) instead of raw torques or position targets. This approach is highly robust and matches our proposed controller design because:
 - it guarantees passivity and safety during learning (since the robot behaves as a passive spring-damper system relative to the rail),
 - it naturally accommodates the low or high stiffness regimes that the exo generates, and
 - it avoids high-frequency corrective torques, leading to smooth, hand-like interactions.

8.2. Alternative Experimental Framing

There are at least three useful ways to frame the experiment.

1. **Human motor-learning experiment:** healthy participants learn the rail–cup–ball task, and we study how stability, predictability, and EMG change with practice.
2. **Robot emulation experiment:** the weak manipulator tries to reproduce healthy hand interaction, while the exo mounted on the manipulator adjusts joint stiffness.
3. **Rehabilitation-inspired transfer experiment:** healthy behavior defines the target, the weak manipulator represents impaired control, and the exo stiffness becomes the teaching signal that the manipulator learns from.

These framings are compatible. The first provides the human reference, the second builds the robot imitation, and the third connects the work to rehabilitation.

9. Linear Rail Experiment Setup Requirements

A good rail setup should be simple enough for ethics approval and repeatable enough for modeling. The safest initial design is a manually actuated two-axis table-top rail with software and mechanical limits, before adding the manipulator-driven version.

Subsystem	Recommended requirement
Rails	Two orthogonal low-friction linear rails with hard mechanical end stops; travel should be large enough for the task but short enough to prevent excessive reach.
Handle	Comfortable handle below the cup, rounded edges, no pinch points, and adjustable height for participant posture.
Cup and ball	Transparent or camera-visible cup, replaceable balls with known mass/radius, and removable target marker inside the cup.
Sensing	6-axis force/torque sensor under the cup, rail encoders or linear scales, overhead camera for calibration, and synchronized EMG.
Disturbance	Solenoid or voice-coil actuator with adjustable amplitude, short pulse duration, physical stroke limit, and software lockout.
Safety	Emergency stop, software travel limits, mechanical stops, current limits, guarded pinch zones, and operator-controlled trial start.
Data	Common timestamp for EMG, force/torque, rail position, camera, auditory feedback, and disturbance trigger.

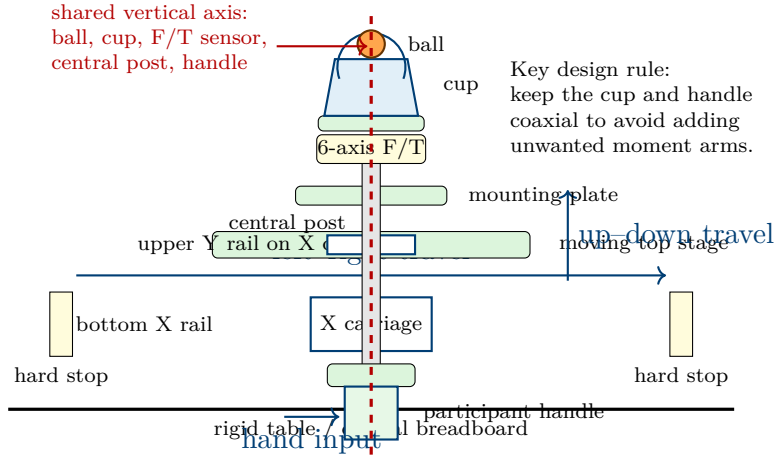


Figure 2: Physical schematic of the two-rail system. The ball, cup, force/torque sensor, central post, and participant handle are mounted on the same vertical axis so that hand input acts directly beneath the cup.

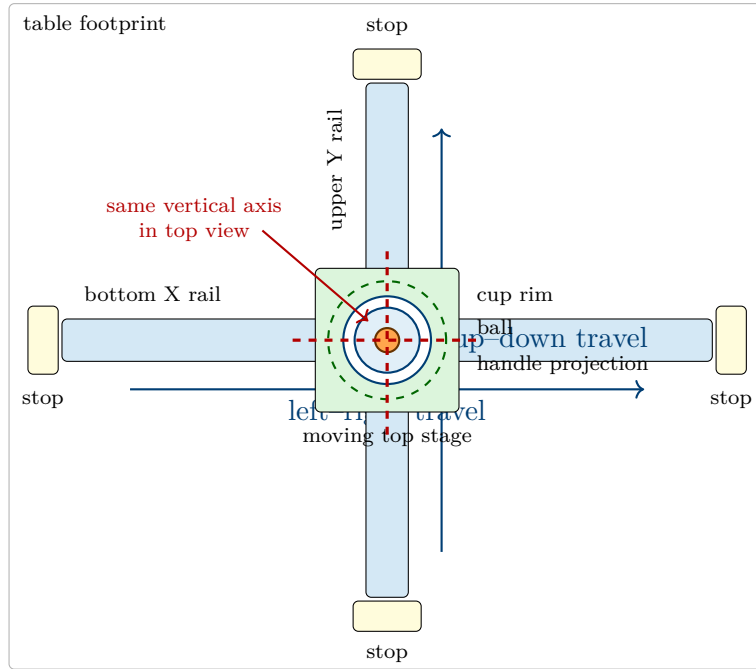


Figure 3: Top view of the two-rail system. The X and Y rail axes cross at the top-stage center, where the ball, cup, force/torque sensor, central post, and handle project to the same point.

10. Ethics and Safety Protocols

For ethics approval, the human rail experiment should be described as a low-risk manual interaction task with controlled perturbations. The key point is that the participant is not strapped to an actuator and can release the handle at any time.

Recommended safety measures:

- Obtain informed consent and explain the disturbance timing, task goals, and right to stop.
- Screen participants for recent upper-limb injury, pain, neurological disorder, severe musculoskeletal limitation, or skin sensitivity to EMG electrodes.

- Use a familiarization block before the ball and before any disturbance.
- Keep the handle passive in the first human experiment; do not rigidly attach the hand or arm.
- Add mechanical end stops and software travel limits on both rail axes.
- Guard all pinch points around rails, pulleys, solenoids, and moving stages.
- Limit disturbance amplitude, duration, and frequency; begin with very small pulses and increase only after pilot safety checks.
- Include an emergency stop accessible to the participant and another to the experimenter.
- Use trial-by-trial verbal check-ins for fatigue, discomfort, dizziness, or pain.
- Stop immediately if the participant reports pain, loses grip, or if sensor readings exceed preset limits.

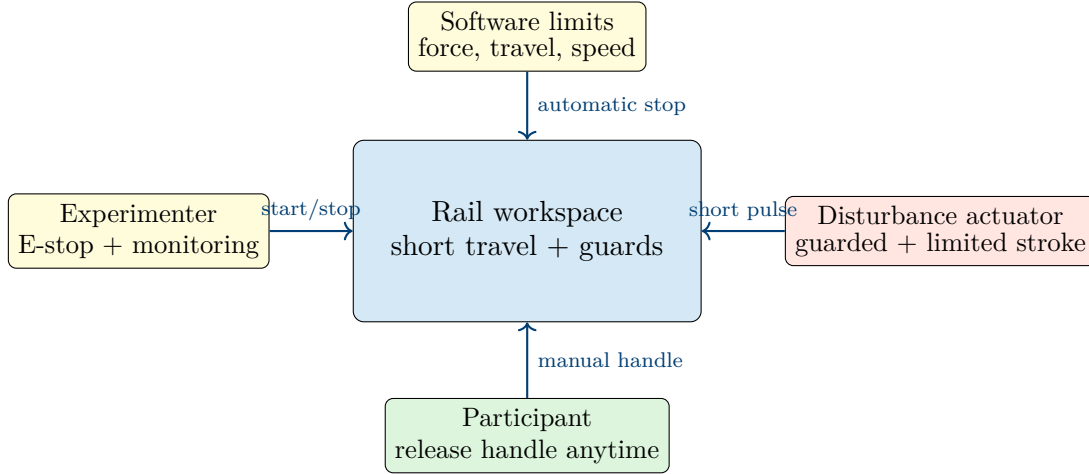


Figure 4: Top-down ethics and safety schematic. The participant interacts through a releasable handle; the actuator is guarded and limited; the experimenter and software both have stop authority.

11. Methodology Adopted from Tele-Impedance (Ajoudani et al., 2012)

Ajoudani et al. (2012) introduced *Tele-Impedance*: a teleoperation scheme in which a slave robot is commanded with *both* a Cartesian position reference *and* a stiffness reference estimated in real time from the agonist–antagonist EMG activity of the human operator. The architectural backbone is a **master–slave pair**: the human operator performs a task on the master side while the robot simultaneously performs the same task on the slave side, with human kinematics and impedance transferred across in real time.

Our experiment uses exactly this architecture:

- **Master side — the two-rail platform.** The human participant performs the ball-in-cup task on the physical rail rig (Figs. 2–3). The participant wears 8-channel surface EMG on the arm that holds the cup handle; a 6-DOF F/T sensor and Optitrack capture handle forces and cup position.
- **Slave side — the SEA cup manipulator with exo-suit.** The 2-DOF cable-driven manipulator reproduces the cup trajectory commanded from the rail side. The agonist–antagonist exo cables simultaneously receive a co-contraction setpoint derived from the participant’s EMG, physically realising the impedance modulation that matches what the human arm is doing on the rail.

This directly mirrors the Ajoudani setup: their human operator caught a ball on the master side; their KUKA LWR caught the same ball on the slave side, receiving position *and* stiffness from the operator’s EMG. We swap the KUKA’s active Cartesian impedance controller for the SEA exo-cable

co-contraction mechanism, and we swap the free-space ball-catching task for the constrained rail ball-in-cup task.

11.1. Master–Slave Architecture

Tele-Impedance — master / slave	Our system — rail / manipulator
<i>Master:</i> human operator performs ball-catch 8-channel arm EMG on operator	<i>Master:</i> participant does ball-in-cup on the 2-rail platform Same 8-channel arm EMG on participant (same muscles, Table I of paper)
Optitrack on wrist / elbow / shoulder Position signal: operator wrist ξ_{human}	Optitrack on cup handle + cart encoder on rail Position signal: cup handle position on rail (x_{cart}, y_{cart})
<i>Slave:</i> KUKA LWR executes same ball-catch Slave stiffness command $\mathbf{K}_{end}$ via KUKA active impedance ctrl	<i>Slave:</i> SEA cup manipulator with exo cables executes same ball-in-cup Slave stiffness via exo cable co-contraction setpoint $c_{exo} \in [0, 1]$ (dual-groove pulley, SEA springs)
6-DOF F/T at slave end-effector	6-DOF F/T at manipulator cup handle; also at rail handle (calibration)
Slave position loop: FR Interface, incremental position	Slave position loop: computed-torque controller on $[q_1, q_2, \dot{q}_1, \dot{q}_2]$ (controller/computed_torque_isaacsim.py)

Data flow during a live trial. At every control cycle (~ 200 Hz):

1. Rail Optitrack/encoder sends cup position (x_c, y_c) to the manipulator position controller.
2. 8-channel EMG from the participant’s arm is preprocessed to envelope $\mathbf{P}$.
3. Null-space projector gives co-contraction signal $\mathbf{P}_{null} = (\mathbf{I} - \hat{\mathbf{T}}^+ \hat{\mathbf{T}}) \mathbf{P}$.
4. Stiffness increment $\Delta \mathbf{K} = \hat{\Psi} \mathbf{P}_{null}$ is converted to an exo co-contraction setpoint via the pulley geometry.
5. Manipulator executes: position from step 1, stiffness from step 4, on top of the SEA torque-mode inner loop.

This preserves the two-channel separation of [Ajoudani et al. \(2012\)](#): position and stiffness are transferred through independent channels, so errors in impedance estimation do not destabilise the position loop.

11.2. Two-Stage Offline Calibration

Calibration is done *on the rail side* (participant at the rail handle, same posture as Block B/C), before the main experiment.

Stage 1 — EMG $\rightarrow$ endpoint-force map $\hat{\mathbf{T}}$. The rail cart is locked; the participant produces isometric forces of $\{5, 10, 20\}$ N along $\pm x, \pm y$ while the rail-handle F/T sensor records ground-truth force. EMG and force are both recorded; least-squares regression solves

$$\hat{\mathbf{T}} = \arg \min_{\mathbf{T}} \sum_j \|\mathbf{F}^{(j)} - \mathbf{T} \mathbf{P}^{(j)}\|^2.$$

Protocol: 4 rep $\times$ 4 dir $\times$ 3 level = 48 trials, 60 s each (first 10 s discarded). Target validation: $R^2 \gtrsim 0.7$ per direction.

Stage 2 — Null-space EMG $\rightarrow$ stiffness map $\hat{\Psi}$. Using $\hat{\mathbf{T}}$, the null-space projector $\mathbf{N} = \mathbf{I} - \hat{\mathbf{T}}^+ \hat{\mathbf{T}}$ isolates the co-contraction signal $\mathbf{P}_{\text{null}}$. The rail cart now applies *stochastic step perturbations* (10–20 mm, 200 ms duration, random xy direction) to the participant’s handle at four TCI levels (min, low, med, high). Endpoint stiffness $\mathbf{K}$ is identified from the perturbation response by fitting a second-order impedance model

$$\mathbf{M}\ddot{\mathbf{X}} + \mathbf{B}\dot{\mathbf{X}} + \mathbf{K}(\mathbf{X} - \mathbf{X}_{eq}) = -\mathbf{F}$$

in the frequency domain (0–10 Hz; $\mathbf{M}$ fixed from the minimum-TCI trial). The linear map $\hat{\Psi}$ is then fitted from $\mathbf{K}_{\text{end}} = \mathbf{K}_{\text{posture}} + \hat{\Psi} \mathbf{P}_{\text{null}}$.

Translating $\hat{\Psi}$ to the exo-cable setpoint. The identified joint stiffness increment $\Delta \mathbf{K}_{\text{joint}}$ (from $\hat{\Psi} \mathbf{P}_{\text{null}}$) is mapped to an exo co-contraction cable pretension via the dual-groove elbow pulley geometry (Method A/B), so that the added SEA spring deflection provides the same stiffness at the manipulator end-effector as the participant’s arm provides at the rail cup handle.

Local validity. As in the original paper, the linear maps $\hat{\mathbf{T}}$ and $\hat{\Psi}$ are valid near the calibration posture. Blocks A–C all operate at the same centred-cart, vertical-cup posture, so one calibration set covers the entire experiment.

11.3. How We Should Proceed With the Experiments

Six-phase plan (rail = master, manipulator = slave):

1. **Phase 0 — Hardware readiness.** (a) Rail: lock/release mechanism, F/T at handle, Optitrack. (b) Exo manipulator: verify exo cable routing, SEA spring values, co-contraction command path in simulation (`script_cup_manipulator_pendulum_tendon_with_exo_pydrake.py`). (c) EMG rig: 8 electrodes (4 flexor / 4 extensor), Delsys or equivalent, 2 kHz, wireless.
2. **Phase 1 — Stage-1 calibration ($\hat{\mathbf{T}}$).** Participant on rail, cart locked. 48 isometric force trials. Fit $\hat{\mathbf{T}}$, report R^2 . Pilot on $N = 2$ before full recruitment.
3. **Phase 2 — Stage-2 calibration ($\hat{\Psi}$).** Rail actuator applies stochastic perturbations at 4 TCI levels. Identify stiffness, fit $\hat{\Psi}$, verify monotone TCI $\uparrow \Rightarrow K \uparrow$. Derive the co-contraction-setpoint lookup for the exo cables.
4. **Phase 3 — Block A (familiarisation, exo passive).** Participant on rail, no ball, no exo stiffness injection. Learns rail dynamics and cup handle feel.
5. **Phase 4 — Block B (live transfer, ball-in-cup).** Within-subject: *Condition 1* exo OFF (manipulator tracks rail position only, no stiffness augmentation); *Condition 2* exo ON EMG-driven (real-time $\hat{\Psi} \mathbf{P}_{\text{null}}$ drives exo co-contraction setpoint on the slave manipulator). Primary metric: ball dwell time / lateral deviation on manipulator.
6. **Phase 5 — Block C (disturbance + adaptation).** External pulse/sine injected at the rail (same perturbation already implemented in *exo-circle-pulse* tasks). Compare disturbance-recovery time and TCI adaptation between conditions. Performance indices analogous to the paper’s BI, PEI, BFR: (a) ball-energy bouncing index from manipulator cup F/T, (b) steady-state ball lateral deviation, (c) ratio of ball excursion before vs. after disturbance.

Key departures from Ajoudani et al. (2012). (1) *Rail replaces free-space human motion as master.* The participant is constrained to the rail; their cup position is the command signal rather than an unconstrained wrist trajectory. (2) *Slave stiffness is mechanical, not active.* Exo SEA cables realise the stiffness increment physically; $\hat{\Psi}$ must be composed with the pulley/spring geometry to recover the cable pretension setpoint (Method A/B, exo-cable instruction file). (3) *2-D planar workspace.* All maps $\mathbb{R}^8 \rightarrow \mathbb{R}^2$ (not $\mathbb{R}^3$) because the rail covers only x and y . (4)

Added Block C. A structured disturbance block probes adaptation, not present in the original paper.

12. Scientific Motivation

The experiment is intended to study how humans learn to control a semi-stable object-platform interaction under increasing complexity.

The block structure is deliberate:

- Block A isolates learning of the rail dynamics,
- Block B introduces stabilization of the ball during voluntary movement,
- Block C adds a disturbance and probes adaptation.

Together with force/torque sensing, camera tracking, and EMG, the setup can provide a detailed view of behavioral and neuromuscular adaptation.

The platform could be useful for studying sensorimotor learning, manual control, disturbance rejection, balance-like stabilization, and co-adaptation of movement and muscle activity.

13. Meeting Summary

- Draft the full experimental protocol for Blocks A, B, and C.
- Define the primary outcome measure for the power analysis.
- Collect a similar paper or pilot data to estimate effect size.
- Plan the camera and force-sensor calibration procedure for ball-position estimation.
- Specify the EMG recording setup and trial structure.
- Prepare an ethics protocol with participant release, end stops, E-stops, actuator limits, and exclusion criteria.
- Build a manual two-axis rail prototype before adding manipulator-driven trials.
- Use Sternad’s complex-object literature to define stability and impedance metrics.

14. Rail-to-Manipulator Transfer: Coordinate Mapping, Weak Stiffness, and Exo-Informed Learning

The core idea: a healthy human teaches a weak robot how to handle an unstable object.

1. The human experiment (master side). A healthy person holds a handle attached to a two-axis rail — one rail moves left-right, the other up-down. On top sits a cup with a ball in it. The person moves the rail while trying to keep the ball near the centre of the cup. This is the “cup and ball” task. Sensors record everything: where the rail goes, how hard they push (force/torque), and how their muscles activate (EMG).

2. Why this is interesting. Keeping the ball stable while moving the cup is not trivial — the ball is an underactuated pendulum-like object. Humans learn to route their motion through trajectories that keep the ball naturally stable, rather than fighting it with brute force. EMG lets us see how their stiffness (muscle co-contraction) changes as they learn.

3. The robot side (slave side). The same task is given to a 2-joint cable-driven robot arm. But deliberately, the robot starts with very weak joint stiffness — it mimics an impaired or unhealthy limb that can’t self-correct well. On its own it would fail the task.

4. The exosuit as teacher. An exosuit (cable-driven elastic harness) is mounted on the robot arm. When the human co-contracts (stiffens up), that EMG signal is read in real time and sent to the exosuit, which physically stiffens the robot’s joints by the same relative amount. So the robot’s stiffness profile now mirrors the human’s — the exo is acting as a stiffness translator from human arm to robot arm.

5. The learning step. While the robot runs with the exo actively shaping its stiffness, data are collected: what state was the robot in, what stiffness did the exo apply, how well did the ball stay on target? A reinforcement learning policy is trained on this data. The reward is: ball stays near centre + robot tracks the rail accurately + don’t use excessive torque.

6. Deploying the learned policy. Once the policy is trained in the digital twin (Phase 2), it is deployed on the real system with the exo hardware still present. The EMG-based co-contraction command is replaced by the policy output: the CT controller continues to track the trajectory while the learned policy regulates the antagonistic cable pre-tension through the elbow exosuit. Experimental evaluation then tests whether this exosuit-mediated stiffness shaping improves tracking stability and disturbance recovery compared to the no-exo (CT-only) baseline.

Three-phase experimental sequence. The overall programme runs in three phases that build on each other:

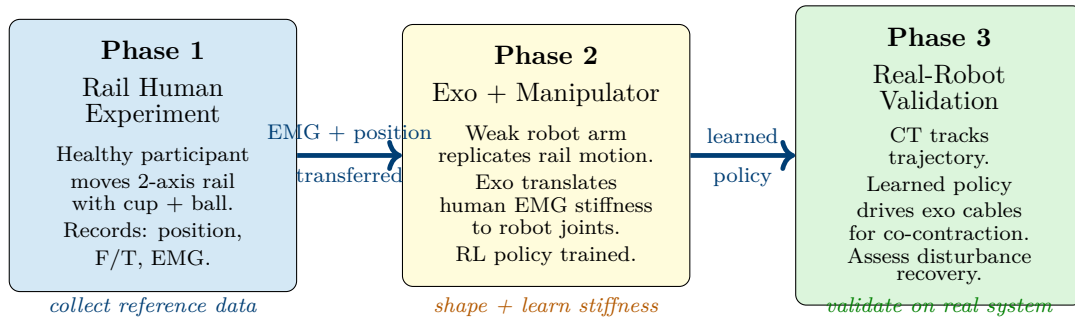


Figure 5: Three-phase experimental sequence. Phase 1 records healthy human behaviour on the rail. Phase 2 builds a digital twin of the manipulator–exosuit and trains an RL policy that reproduces the desired Cartesian impedance. Phase 3 deploys the learned policy on the real robot: the CT controller handles trajectory tracking and the learned policy drives the exo cables to produce co-contraction and stiffness modulation.

This section formalises the three-step technical pipeline that connects the human rail experiment to the simulated exosuit manipulator.

14.1. Phase-Wise Protocol and Data Mapping

14.1.1 Phase 1 — Human Rail Experiment (Master Side)

Objective. Collect high-quality reference behaviour from healthy participants while they perform the rail cup-and-ball task.

Operational steps.

1. Run Block A (familiarisation, no ball) to learn rail dynamics and timing.
2. Run Block B (ball-in-cup tracking) to capture nominal stabilisation behaviour.
3. Run Block C (disturbance adaptation) to capture corrective and recovery behaviour.
4. Record all streams with a common timestamp: rail position, F/T, EMG, camera-based ball state, disturbance trigger.
5. Perform quality checks per trial (signal dropout, clipping, tracking loss, sync quality).

Block definitions inside Phase 1.

Block A: Familiarisation without the ball. In Block A, participants complete 10 trials without the ball. The aim is to let them learn the platform dynamics and become familiar with how hand motion affects the two-rail system.

Participants are asked to move the platform left to right, right to left, and up and down within a specified time window. Feedback is auditory: a pleasant sound for successful completion and an unpleasant sound for failure.



Figure 6: Phase-1 Block A schematic. Participants learn the rail motion without the ball so the platform dynamics can be learned first.

Block B: Ball-in-cup control task. In Block B, the ball is added to the cup. A circular target region is defined, and participants move the platform while keeping the ball inside the target as much as possible.

Success can be defined by one or more criteria: completion within the time limit, ball remaining inside the target circle, and boundary violations below a threshold.

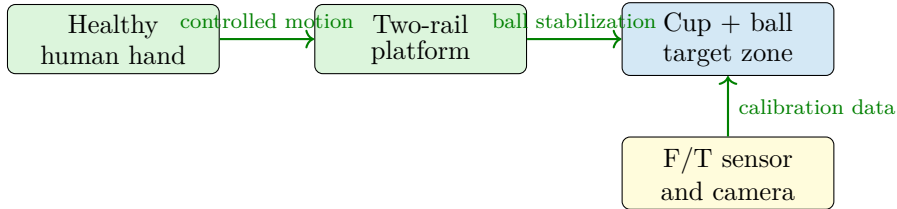


Figure 7: Phase-1 Block B schematic. The ball must stay near the target while the participant moves the rail platform.

Block C: Disturbance adaptation. In Block C, an external disturbance is introduced using a solenoid actuator that pushes the rail toward the participant. The perturbation disturbs platform motion and moves the ball away from the desired region.

Participants adapt their control strategy over repeated trials to bring the ball back to target. This block probes disturbance rejection, error-based motor learning, and recovery dynamics.

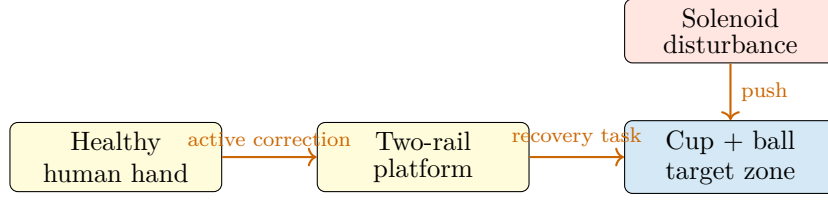


Figure 8: Phase-1 Block C schematic. A disturbance pushes the rail, and the participant learns to recover the ball back to the target.

Rail setup diagrams for Phase 1. The physical and top-view rail geometries used in Phase 1 are shown in Figs. 2 and 3. These diagrams define the coordinate axes, coaxial cup-handle mounting, and sensor placements used for data collection and calibration.

Data products from Phase 1.

- Trajectory reference: $(x_c(t), y_c(t))$
- Interaction force reference: $\mathbf{F}(t)$
- Muscle activation/co-contraction features: $\mathbf{P}(t)$, $\mathbf{P}_{\text{null}}(t)$
- Ball-state performance labels: dwell time, boundary violations, recovery time

14.1.2 Phase 2 — Exo + Manipulator (Digital Twin and RL)

Objective. Build a digital twin of the manipulator–exosuit system that captures robot dynamics, cable transmission, SEA compliance, and elbow stiffness modulation. Train an RL policy in the digital twin to reproduce the desired Cartesian impedance behaviour from the Phase 1 human demonstrations.

Operational steps.

1. Feed rail position (x_c, y_c) into IK to generate joint references $\mathbf{q}^*(t)$.
2. Run weak-baseline CT tracking on the manipulator.
3. Convert EMG co-contraction feature $\mathbf{P}_{\text{null}}$ to stiffness increment via $\hat{\Psi}$ and map to exo command c_{exo} .
4. Apply exo cable co-contraction in real time; this adds mechanical stiffness to the weak arm.
5. Collect closed-loop robot rollout data and train policy (imitation pretraining + RL fine tuning).

Data products from Phase 2.

- Robot state-action trajectories $(\mathbf{s}_t, \mathbf{a}_t)$
- Exo stiffness command trajectory $c_{\text{exo}}(t)$
- Task outcomes under assisted stiffness (ball error, tracking error, effort)
- Policy parameters π_θ and validation metrics

14.1.3 Phase 3 — Real-Robot Validation (Learned Policy via Exo)

Objective. Deploy the trained RL policy on the physical manipulator–exosuit and validate whether exosuit-mediated stiffness shaping improves tracking stability and recovery under real-world disturbances.

Operational steps.

1. Connect the learned policy π_θ to the exo co-contraction command path, replacing the EMG-based Tele-Impedance signal from Phase 2.
2. The CT controller continues to handle end-point trajectory tracking independently.
3. The learned policy generates the antagonistic cable pre-tension setpoint $c_{\text{exo}}(t)$, physically producing co-contraction and stiffness modulation through the elbow exosuit.

4. Run validation trials: nominal ball-in-cup tracking and disturbance recovery (same perturbation protocol as Phase 1 Block C).
5. Compare Exo OFF (CT only) against Exo ON (CT + learned policy) to quantify the benefit of exosuit-mediated stiffness shaping.

Success criteria.

- Ball stabilisation metrics under the learned policy match or exceed Phase 2 (EMG-driven) performance.
- Disturbance recovery is faster and more repeatable with the learned policy driving the exo than without.
- The co-contraction profile is interpretable: stiffness increases near perturbations.
- Cable tensions and control effort remain within safe operational limits.

14.1.4 Cross-Phase Data-to-Use Mapping

Signal (Phase 1)	Used in Phase 2	Used in Phase 3
Rail position (x_c, y_c)	IK reference for $\mathbf{q}^*(t)$	Same task reference for final validation
EMG envelope $\mathbf{P}$	Co-contraction extraction and exo command	Policy feature or distilled gain schedule input
F/T at handle	Calibration of $\hat{\mathbf{T}}$ and stiffness identification	Benchmark for interaction similarity
Ball state (camera/F/T map)	RL reward and trial scoring	Final autonomous performance scoring
Disturbance timing/amplitude	Robustness training episodes	Robustness validation episodes

14.2. Step 1 — Coordinate Mapping: Rail Cart to Manipulator End-Effector

The two-rail platform defines a 2-D Cartesian position $\mathbf{p}_{\text{rail}}(t) = [x_c(t), y_c(t)]^\top \in \mathbb{R}^2$, measured by the rail encoders or Optitrack. The manipulator end-effector (cup tip) must follow the same trajectory in the workspace.

Forward kinematics (FK). Let $\mathbf{q} = [q_1, q_2]^\top$ be the shoulder and elbow joint angles of the 2-DOF cable-driven manipulator (link lengths L_1, L_2). The cup tip position in the world frame is

$$p_x = L_1 \cos q_1 + L_2 \cos(q_1 + q_2), \quad (3)$$

$$p_y = L_1 \sin q_1 + L_2 \sin(q_1 + q_2). \quad (4)$$

Inverse kinematics (IK). Given $\mathbf{p}_{\text{ref}}(t) = \mathbf{p}_{\text{rail}}(t)$ the required joint trajectory is obtained by solving

$$\mathbf{q}^*(t) = \arg \min_{\mathbf{q}} \|\mathbf{p}(\mathbf{q}) - \mathbf{p}_{\text{ref}}(t)\|^2,$$

subject to joint limits. In the simulation this is solved numerically at each trajectory sample using Drake’s `InverseKinematics` class with warm-starting (previous solution used as seed). The rail workspace is chosen so that the two-link arm can cover the full travel without reaching a singular configuration.

Practical workflow in simulation.

1. Rail encoder/Optitrack streams $\mathbf{p}_{\text{rail}}(t)$ at the control rate (~ 200 Hz).

2. IK solves for $\mathbf{q}^*(t)$; the result is the position setpoint.
3. The computed-torque controller (`controller/computed_torque_isaacsim.py`) tracks $\mathbf{q}^*(t)$ on the simulated manipulator.
4. The SEA exo cables simultaneously receive the co-contraction setpoint derived from EMG (Section 11).

Joint-order note (Drake). Drake returns joint indices in URDF parse order, not alphabetical. For `cup_manipulator.urdf` the ordering is `[link2_link1, link1_base]`, i.e. the state vector is $[q_2, q_1]^\top$. Always verify with `plant.GetJointIndices()` before setting IK seeds or reading solutions.

14.3. Step 2 — Weak-Stiffness Manipulator Model

The manipulator is intentionally operated with *reduced* joint stiffness to emulate an impaired or unhealthy limb. In the computed-torque (CT) framework the desired joint torque is

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}}^* + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + K_p^{\text{eff}}(\mathbf{q}^* - \mathbf{q}) + K_d^{\text{eff}}(\dot{\mathbf{q}}^* - \dot{\mathbf{q}}), \quad (5)$$

where the *effective* PD gains are split into a weak baseline and an exo-augmented increment:

$$K_p^{\text{eff}} = K_p^{\text{weak}} + \Delta K_p^{\text{exo}}, \quad (6)$$

$$K_d^{\text{eff}} = K_d^{\text{weak}} + \Delta K_d^{\text{exo}}. \quad (7)$$

The weak-baseline gains satisfy $K_p^{\text{weak}} = \varepsilon K_p^{\text{healthy}}$ with $\varepsilon \ll 1$ (typical range $\varepsilon \in [0.1, 0.3]$), and the exo increment $\Delta K_p^{\text{exo}} \geq 0$ is contributed by the exo-cable co-contraction (dual-groove elbow pulley, Method A/B).

At the SEA level the exo cable pretension T_c adds to the joint stiffness as

$$\Delta K_p^{\text{exo}} \approx \frac{r_p^2 k_s}{1} c_{\text{exo}}, \quad (8)$$

where r_p is the pulley radius, k_s is the SEA spring stiffness, and $c_{\text{exo}} \in [0, 1]$ is the dimensionless co-contraction command. This relation is the link between the Tele-Impedance stiffness increment $\Delta \mathbf{K}$ (identified in Stage-2 calibration) and the physical cable setpoint.

Mode	K_p^{weak}	ΔK_p^{exo}	Description
Healthy baseline	K_p^{healthy}	0	full CT, no exo
Weak (impaired)	$\varepsilon K_p^{\text{healthy}}$	0	emulates impaired control
Weak + exo OFF	$\varepsilon K_p^{\text{healthy}}$	0	rail EMG not yet connected
Weak + exo ON	$\varepsilon K_p^{\text{healthy}}$	$\Delta K_p^{\text{exo}}(t)$	live EMG-driven co-contraction
Learned	$K_p^{\text{learned}}(t)$	0	exo distilled into CT gains

14.4. Step 3 — Exo-Informed Stiffness Learning

The long-term goal is for the manipulator to *inherit* the stiffness modulation that the exo currently provides, so the exo can eventually be removed and the manipulator controller reproduces the same interaction behavior autonomously.

Learning formulation. Define the state $\mathbf{s}(t) = [\mathbf{q}, \dot{\mathbf{q}}, \mathbf{p}_{\text{ball}}, \dot{\mathbf{p}}_{\text{ball}}, \mathbf{p}_{\text{rail}}]$ and the action $\mathbf{a}(t) = [c_{\text{exo}}, \boldsymbol{\tau}_{\text{ff}}]$ (exo co-contraction setpoint plus optional feed-forward torque adjustment). The RL policy

$$\mathbf{a}(t) = \pi_{\theta}(\mathbf{s}(t))$$

is trained to minimise a cost function that rewards ball retention and smooth rail tracking:

$$\mathcal{J} = \mathbb{E} \left[\int_0^T \underbrace{w_1 \|\mathbf{p}_{\text{ball}} - \mathbf{p}_{\text{target}}\|^2}_{\text{ball position error}} + \underbrace{w_2 \|\mathbf{p}_{\text{EE}} - \mathbf{p}_{\text{rail}}\|^2}_{\text{EE tracking error}} + \underbrace{w_3 \|\boldsymbol{\tau}\|^2}_{\text{effort penalty}} + \underbrace{w_4 \|\mathbf{K}_{\text{eff}}(t) - \mathbf{K}_{\text{ref}}(t)\|_F^2}_{\text{impedance matching}} dt \right], \quad (9)$$

The fourth term rewards matching the Cartesian stiffness reference $\mathbf{K}_{\text{ref}}(t)$ estimated from Phase 1 human EMG and posture data, which encourages the policy to produce co-contraction that mirrors the human impedance response—in particular increasing c_{exo} near perturbations.

Data collected during *Weak + exo ON* trials (where the exo acts as a teacher) provide the demonstration set for imitation-learning pre-training; PPO fine-tunes the policy thereafter.

Policy deployment through the exosuit (Phase 3). Once π_{θ} is trained in the digital twin, it is deployed on the real manipulator–exosuit with the exo hardware still present. The EMG-based Tele-Impedance co-contraction command is replaced by the policy output:

$$c_{\text{exo}}(t) = \pi_{\theta}^{(c)}(\mathbf{s}(t)), \quad (10)$$

where $\pi_{\theta}^{(c)}$ is the co-contraction output channel of the trained policy and $\mathbf{s}(t)$ contains robot joint states, end-effector motion, and interaction information. The CT controller continues to track the end-point trajectory independently. This two-channel architecture (position via CT, stiffness via learned exo policy) mirrors the Tele-Impedance master–slave structure of Section 11 but replaces the human operator’s EMG with an autonomous policy. Experimental evaluation then assesses whether exosuit-mediated stiffness shaping improves tracking stability and disturbance recovery relative to the no-exo (CT-only) baseline.

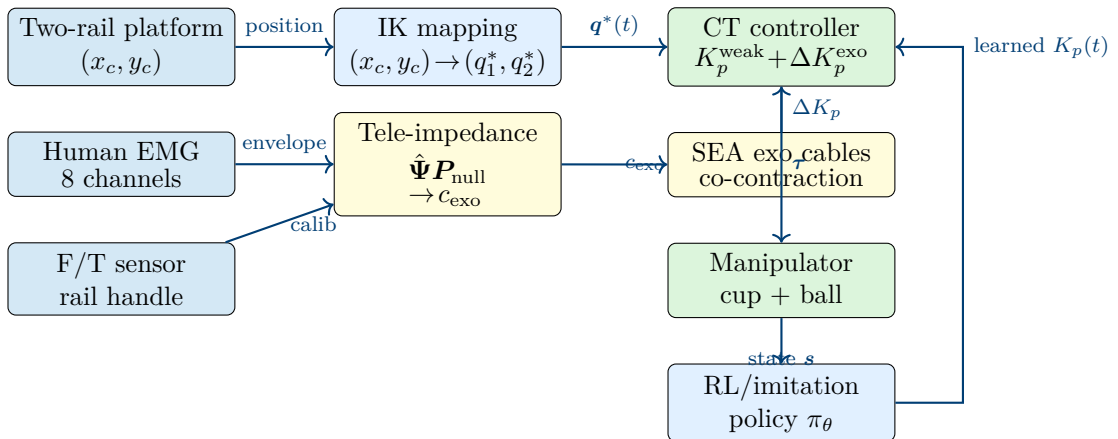


Figure 9: Full rail-to-manipulator transfer pipeline. Left: human master side (rail, EMG, F/T). Right: robotic slave side (CT controller, SEA exo, manipulator cup + ball). The RL policy closes the loop by distilling exo-shaped stiffness directly into the CT gain schedule, so the exo can eventually be removed.

Simulation entry points.

- Exo co-contraction PyDrake simulation: `script_cup_manipulator_pendulam_tendon_with_exo_pydra`
- Isaac Sim version: `script_cup_manipulator_pendulam_tendon_with_exo_isaac_sim.py`
- CT controller (engine-agnostic): `controller/computed_torque_isaacsim.py`
- SEA cable model: `actuators/sea_isaacsim.py`
- RL training: `rl/train_ppo_residual.py`

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